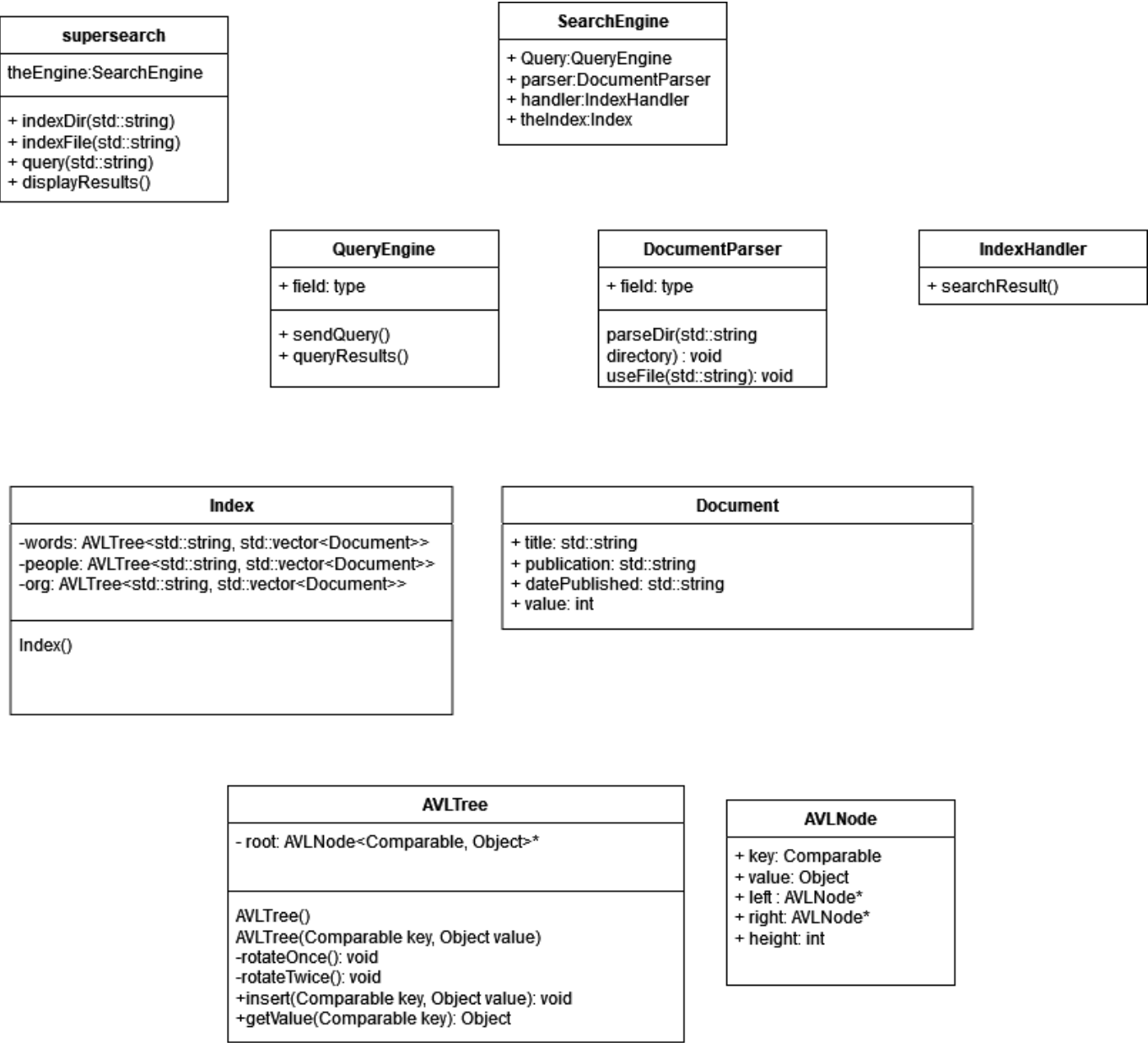


UML Diagram



Description of Classes

supersearch: My UI implementation and what will be used in the command line to use application. This is a basic driver.

QueryEngine: Used to strip queries to their stams and remove stop words from queries. Will also return search results to SuperSearch UI.

DocumentParser: Parses file in directory for index or updates index using persistent index file

Index: Holds the AVL Trees (or inverted file index) for words, people, and orgs used to generate search results

IndexHandler: Determines what is relevent using index class

SearchEngine: Object that instantiates classes for ease of use by driver/UI.

Document: Contains document information for later display

AVLTree & AVLNode: AVL implementation

Class interaction

- **supersearch** gives **DocumentParser** dir of files to parse or persistent index file to use
- **DocumentParser** works with **Index** to make inverted file indexes for words, people, and orgs which are really AVL Tree data structures that act as maps with values consisting of std::vectors of **Document**. During this process stemming and stop word removal is done to words.
- **supersearch** sends query to **QueryEngine**, which does a similar process to **DocumentParser** in stripping the query search of stop words and stemming the words.
- **QueryEngine** sends stripped query to **IndexHandler**, which uses **Index**'s inverted file index's to generate a search result based of some sort of relevency metric.
- **IndexHandler** returns search result to **QueryEngine** which in turn returns the result to **supersearch** for display to the user.

Schedule

Task	Planned completion	Actual completion	Time spent in hrs
AVL Tree Implementation and Catch 2 Tests. Finish UML Design	April 10		
Implement the document parser; build the indices.	April 13		
Implement the query processor; user interface; and add persistence to the index.	April 20		
Everything else	April 27		